

Tighter error bounds using structural floating-point properties and tail bounds

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Abstract—Traditional deterministic error analysis provides worst case error bounds, and prior work on probabilistic error analysis do not account for structural floating-point properties, leading to looser bounds for cancellation heavy cases. In this work, we use hybrid error analysis to provide tighter error bounds for long summation and dot product numerical kernels. We extend the FPChecker tool to track both global and local rounding error, which helps in computing key statistics related to structural floating-point properties, such as the Sterbenz lemma. We then use these computed statistics in tail bounds for rounding error which provide average case certificate or probabilistic error guarantees. We also provide modified Bernstein’s tail bound by incorporating the dependency graph approach first provided by Janson et al., to model correlation between local error at different steps. We then compare our derived error bound with Higham and Mary’s probabilistic error bound and show that our bound is 2-3 orders of magnitude tighter for cancellation heavy cases due to incorporation of the structural floating-point behavior in our error model. Finally, we show the applicability of the derived bounds on a physical example of computing the magnitude of Fourier modes for density wave function used in compressible-flow simulations.

Index Terms—Tail bounds, Stochastic rounding, Floating-point error, Probabilistic error analysis, Hybrid error analysis

I. INTRODUCTION

Traditional error analysis [1] gives worst case error bounds. For operations such as summation and dot product, the error bounds are in the order of $\mathcal{O}(nu)$, where n is the size of the vector and u is the unit round-off error. These bounds do not take into account structural floating-point properties, i.e., cases where floating-point operations are exact such as in the Sterbenz lemma [2] (subtraction of two floating-point numbers a and b is exact if $|a|/2 \leq b \leq 2|a|$), and therefore fail to capture precise numerical behavior leading to pessimistic bounds.

To overcome the limitations of deterministic error analysis, several works [3]–[11] have demonstrated the effectiveness of *probabilistic* floating-point error analysis for providing tighter bounds by modeling rounding errors as random variables to capture cancellation effects during successive operations.

These works provide tail bounds which bound the tail behavior of the actual error distribution with some confidence ζ . For instance, $\zeta = 0.95$ provides a 95% probability bound, which means that the computed rounding error will be within the derived bound with 95% probability. Choice of tail bound determines how tight the bounds are. For instance, Higham and Mary [6] use Hoeffding’s concentration inequality [12] to derive tight bound for summation and dot product exhibiting $\sqrt{n}$ growth instead of n growth from deterministic analysis. Tao et al. [11] use Markov’s inequality to analyze arithmetic expressions efficiently. Bhola and Duraisamy [3] use Bernstein’s concentration inequality [12] to derive tighter bounds for summation and dot product. The tightness of error bounds derived from Bernstein’s inequality comes from the fact that Bernstein’s bound takes into account the first (mean) and second order (variance) moments of the error distribution, but Hoeffding’s inequality takes into account only the first moment (mean) of the error distribution. However, these work still fail to capture precise numerical behavior of floating-point operations and do not accurately track the cancellation of rounding errors (see Fig. 4b in Bhola and Duraisamy [3]). We further reinforce this point using a pathological example of long summation $\sum_k x_k$, where each element of x_k is drawn from a uniform distribution and has alternate signs, i.e. x_k is positive when k is even and x_k is negative when k is odd. Figure 1 illustrates that for such a case even $\sqrt{n}u$ bound is roughly four orders of magnitude higher than the actual rounding error. This is due to the fact that standard statistical bounds do not account for the actual numerical behavior at runtime.

In this work, we introduce hybrid analysis, where we profile the numerical behavior at runtime using the FPChecker [13] instrumenter to capture key statistics (discussed in Section IV), and provide an average case certificate by deriving a probabilistic error bound similar to Higham and Mary [6], but parameterized with the computed statistics. We then evaluate our derived error bounds with respect to the Higham-Mary bound [6] and actual rounding error for long summation and

dot product numerical kernels, and show that we can get tighter bounds within two orders of magnitude of the actual rounding error by incorporating runtime numerical behavior. Finally, we show the applicability of the derived bounds on a physical example of computing the magnitude of Fourier modes for density wave function used in compressible-flow simulations, which involves numerical integration using quadrature (long-summation kernel) with symmetric input interval.

Our key contributions are:

- the extension of FPChecker with error tracking mechanism;
- the mixture of error models which capture the structural floating-point properties and incorporates profiled numerical behavior from FPChecker;
- the derivation of modified Bernstein's tail bound which relaxes the condition on independence between local errors by incorporating dependency graph approach from Janson et al [14].

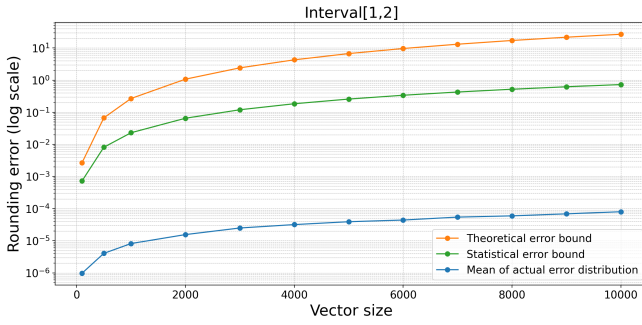


Fig. 1. Error plot for $\sum_k x_k$ in $[1, 2]$; $x_k > 0$ when k is even and $x_k < 0$, when k is odd.

II. AVERAGE CASE CERTIFICATES USING PROBABILISTIC ANALYSIS

To derive our probabilistic error bound informed by structural properties of floating-point operations (discussed in Section IV), we start with Bernstein's concentration inequality (similar to Bhola and Duraisamy [3]), which is stated as follows.

Theorem 2.1 (Bernstein's inequality): Let $\{Z_i\}_{i=1}^n$ be n independent random variables such that $|Z_i| \leq c$ almost surely, for all i . Then, the random variable $S_n := \sum_{i=1}^n Z_i$ satisfies

$$P(|S_n - \mathbb{E}[S_n]| \geq t) \leq 2 \exp \left(-\frac{t^2}{2(\sigma^2 + ct/3)} \right) \quad (1)$$

for all positive t , where $\sigma^2 := \sum_{i=1}^n \text{Var}[Z_i]$.

Here, c is called the almost sure bound of Z_i . We assume that the mean of S_n , $\mathbb{E}[S_n] = 0$. To compute a high probability bound on $|S_n|$ with probability $\geq 1 - \delta$, we choose t so that RHS of 1 is $\leq \delta$.

$$\begin{aligned} 2 \exp \left(\frac{-t^2}{2V + \frac{2}{3}ct} \right) &\leq \delta \\ \Rightarrow \frac{-t^2}{2V + \frac{2}{3}ct} &\leq \ln(\delta/2); \text{ pose } L := \ln(2/\delta) \\ \Rightarrow \frac{t^2}{2V + \frac{2}{3}ct} &\geq L \\ \Rightarrow t^2 &\geq L \left(2V + \frac{2}{3}ct \right) \\ \Rightarrow t^2 - \frac{2}{3}cLt - 2VL &\geq 0 \\ \Rightarrow t &\geq \frac{\frac{2}{3}cL + \sqrt{\left(\frac{2}{3}cL\right)^2 + 8VL}}{2} \end{aligned}$$

Simplifying using the inequality $\sqrt{x+y} \leq \sqrt{x} + \sqrt{y}$; $x, y \geq 0$, we get $t \geq \sqrt{2VL} + \frac{2}{3}cL$. Hence,

$$|S_n| \leq \sqrt{2VL} + \frac{2}{3}cL; \quad L = \ln(2/\delta) \quad (2)$$

For a 95% probability bound $\delta = 0.05$. Therefore, to compute a high probability bound for $|S_n|$ from equation 2, we need to compute two parameters: (a) the almost sure bound, c for $|Z_i|$; and (b) the sum of variances V , of the actual distributions Z_i . For summation and dot product numerical kernels, we show the derivation of these parameters for the assumed distribution of Z_i in Section IV.

However, Bernstein's concentration inequality assumes independence between the random variables Z_i , which implies that $\text{Var}(\sum_i Z_i) = \sum_i \text{Var}(Z_i) + 2 \sum_{i < j} \text{Cov}(Z_i, Z_j)$ simplifies to $\sum_i \text{Var}(Z_i)$ since $\sum_{j < i} \text{Cov}(Z_i, Z_j) = 0$. This assumption about independence of Z_i and Z_j is often too strict and may not represent the actual error behavior. In principle, since the rounding error is propagated forward, Z_i depends on all Z_j for $j < i$. Janson et al [14] derived bounds for summation for partly dependent random variables. Their approach builds a dependency graph for Z_i with a maximum degree of dependency d . They derived a modified Hoeffding's bound (Theorem 2.1 in their paper [14]), which is given by $P(X \geq \mathbb{E}[X] + t) \leq \exp \left(-2 \frac{t^2}{\chi^*(\mathcal{A}) \sum_{\alpha \in \mathcal{A}} (b_\alpha - a_\alpha)^2} \right) \leq \exp \left(-2 \frac{t^2}{(d+1) \sum_{\alpha \in \mathcal{A}} (b_\alpha - a_\alpha)^2} \right)$, where $a_\alpha \leq Y_\alpha \leq b_\alpha$, and $\chi^*(\mathcal{A}) \leq (d+1)$.

We next provide a derivation of the modified Bernstein's bound following the arguments of Janson et al. [14] in their derivation of Hoeffding's bound.

A. Derivation of modified Bernstein's bound

1) Setup and Structural Definitions: Let $X_1, X_2, \dots, X_N$ be centered random variables ($\mathbb{E}[X_k] = 0$) bounded by M , meaning $|X_k| \leq M$ almost surely. Let $\mathcal{G} = (V, E)$ be a dependency graph for these variables, where vertices $V = \{1, \dots, N\}$. If there is no edge between index i and index j , then X_i and X_j are independent. The maximum degree of any vertex in this graph is denoted by d . By definition of graph coloring, the vertices of any graph with maximum degree d

can be colored using at most $\chi(\mathcal{G}) \leq d+1$ colors. This means we can partition our index set V into $d+1$ disjoint subsets (color classes):

$$V = I_1 \cup I_2 \cup \dots \cup I_{d+1}$$

Because no two adjacent vertices share the same color, all random variables within any single subset I_m are mutually independent.

2) *Decomposing the Goal:* We want to bound the tail probability of the total sum $S_N = \sum_{k=1}^N X_k$. We split this total sum into $d+1$ independent sub-sums:

$$S_N = \sum_{m=1}^{d+1} S(I_m), \quad \text{where } S(I_m) = \sum_{k \in I_m} X_k$$

Using Chernoff's bounding method for any parameter $\lambda > 0$:

$$\begin{aligned} \mathbb{P}(S_N \geq T) &= \mathbb{P}(\exp(\lambda S_N) \geq \exp(\lambda T)) \\ &\leq \exp(-\lambda T) \mathbb{E} \left[\exp \left(\lambda \sum_{m=1}^{d+1} S(I_m) \right) \right] \end{aligned}$$

3) *Applying Hölder's Inequality:* Since the sub-sums are inside a single exponential, they are coupled. We decouple them using a generalized Hölder's Inequality with equal weights $p_m = d+1$:

$$\mathbb{E} \left[\exp \left(\lambda \sum_{m=1}^{d+1} S(I_m) \right) \right] \leq \prod_{m=1}^{d+1} (\mathbb{E} [\exp(\lambda(d+1)S(I_m))])^{\frac{1}{d+1}}$$

4) *Bounding the Independent Sub-Sums:* Now we examine a single term in the product. Because all X_k for $k \in I_m$ are mutually independent, the expectation of their exponential sum factors into a product of expectations:

$$\mathbb{E} [\exp(\lambda(d+1)S(I_m))] = \prod_{k \in I_m} \mathbb{E} [\exp(\lambda(d+1)X_k)]$$

We now apply the standard Bernstein lemma expansion for a single bounded, centered random variable X_k . For any $|t| \leq 3/M$:

$$\mathbb{E}[\exp(tX_k)] \leq \exp \left(\frac{t^2 \text{Var}(X_k)}{2(1 - |t|M/3)} \right)$$

Substituting $t = \lambda(d+1)$, we require $\lambda(d+1) < 3/M$:

$$\mathbb{E} [\exp(\lambda(d+1)X_k)] \leq \exp \left(\frac{\lambda^2(d+1)^2 \text{Var}(X_k)}{2(1 - \lambda(d+1)M/3)} \right)$$

Multiplying these terms back together over the index set I_m :

$$\mathbb{E} [\exp(\lambda(d+1)S(I_m))] \leq \exp \left(\frac{\lambda^2(d+1)^2 \sum_{k \in I_m} \text{Var}(X_k)}{2(1 - \lambda(d+1)M/3)} \right)$$

5) *Reassembling the Total Sum:* Now, we substitute this back into our Hölder product expression from Step 3:

$$\mathbb{E} [\exp(\lambda S_N)] \leq \prod_{m=1}^{d+1} \left[\exp \left(\frac{\lambda^2(d+1)^2 \sum_{k \in I_m} \text{Var}(X_k)}{2(1 - \lambda(d+1)M/3)} \right) \right]^{\frac{1}{d+1}}$$

Distributing the exponent $\frac{1}{d+1}$ eliminates one degree of the polynomial:

$$\mathbb{E} [\exp(\lambda S_N)] \leq \prod_{m=1}^{d+1} \exp \left(\frac{\lambda^2(d+1) \sum_{k \in I_m} \text{Var}(X_k)}{2(1 - \lambda(d+1)M/3)} \right)$$

Since the subsets I_m partition the entire vertex set V , summing $\sum_{k \in I_m} \text{Var}(X_k)$ across all m yields the sum of independent variances over the entire sequence: $\sum_{k=1}^N \text{Var}(X_k) = \sum_{k=1}^N \text{Var}(X_k)$.

$$\mathbb{E} [\exp(\lambda S_N)] \leq \exp \left(\frac{\lambda^2(d+1) \sum_{ind}^2}{2(1 - \lambda(d+1)M/3)} \right)$$

6) *Optimizing λ :* Returning to our Chernoff bound from Step 2, we have:

$$\mathbb{P}(S_N \geq T) \leq \exp \left(-\lambda T + \frac{\lambda^2(d+1) \sum_{ind}^2}{2(1 - \lambda(d+1)M/3)} \right)$$

To simplify the calculation, we make a change of variable: let $\theta = \lambda(d+1)$. The bound becomes:

$$\mathbb{P}(S_N \geq T) \leq \exp \left(-\frac{\theta T}{d+1} + \frac{\theta^2 \sum_{ind}^2}{2(d+1)(1 - \theta M/3)} \right)$$

We optimize this by choosing the standard Bernstein parameter $\theta = \frac{T}{\sum_{ind}^2 + MT/3}$. Substituting this specific value of θ back into the exponent and simplifying yields:

$$\mathbb{P}(S_N \geq T) \leq \exp \left(-\frac{T^2/2}{(d+1)(\sum_{ind}^2 + \frac{MT}{3})} \right) \quad (3)$$

7) *Modified Bernstein's bound:* With the modified Bernstein's concentration inequality 3, Bernstein's bound 2 modified to

$$|S_n| \leq \sqrt{2(d+1)VL} + \frac{2}{3}(d+1)cL \quad \text{with } L = \ln(2/\delta) \quad (4)$$

III. ERROR TRACKING IN FPCHECKER

FPChecker dynamically computes line-level rounding error by combining compilation-time LLVM instrumentation with a lightweight runtime. In practice, the application is compiled with a runtime header and an LLVM pass plugin. The pass injects callbacks around floating-point additions, subtractions, multiplications, divisions, fused multiply-adds, selected math-library calls, and the data-movement operations needed to preserve provenance across loads, stores, control-flow joins, and function boundaries. Because each callback carries the source filename and line number, every floating-point event observed at runtime can be mapped back to the exact statement that produced it.

At runtime, FPChecker compares an isolated single-precision result against a fresh double-precision recomputation. It also propagates previously accumulated error information through the dataflow. For an instrumented floating-point instruction, a runtime routine reconstructs a high-precision value from the operands after adding their propagated errors, then compares that reference with the actual FP32 result produced by the program.

For this paper, we instrument a program in FP32 precision, and use FP64 precision as shadow operations to calculate the absolute and relative error of each operations. The absolute rounding error is modeled as

$$\varepsilon_{\text{abs}} = r_{64} - r_{32},$$

while the relative error is computed as $\varepsilon_{\text{rel}} = |\varepsilon_{\text{abs}}|/|r_{64}|$ whenever the high-precision reference is numerically meaningful; if the reference is effectively zero, FPChecker avoids an

unstable normalization and records an infinite relative error unless the absolute error is exactly zero. This detail is important for later analyses that distinguish genuinely exact operations from operations whose relative error is ill-conditioned.

A. Modeling Global and Local Error

The propagation mechanism is implemented through *global runtime tables* that mirror the program state at the granularity needed by the instrumentation. One hash table stores the current error associated with floating-point LLVM registers, keyed by register name and function name, and a second hash table stores the error associated with memory addresses so that values preserve their error metadata when they are written to or read from memory. Each entry records the absolute error, the relative error, the source location, and a logical clock that identifies the most recent update. Additional hooks handle PHI nodes by using the last executed basic block to choose the incoming value, copy error metadata across `memcpy/memmove`, and transfer errors through function calls by pushing argument and return-value information across call boundaries.

For local rounding error, the error is computed at each operation, i.e., the operands do not carry the propagated error. We do not reconstruct shadow values, r_{64} . Hence, given an operands x_{32} and y_{32} , operation op , we compute the local rounding error for op as

$$\epsilon_{local} = (x_{32 \rightarrow 64} \text{ } op_{64} \text{ } y_{32 \rightarrow 64}) - (x_{32} \text{ } op_{32} \text{ } y_{32})$$

$x_{32 \rightarrow 64}, y_{32 \rightarrow 64}$ simply cast x_{32} and y_{32} to double precision, incurring no rounding error. The global error on the other hand would be

$$\epsilon_{global} = ((x_{32} + \epsilon_{x,prop}) \text{ } op_{64} \text{ } (y_{32} + \epsilon_{y,prop})) - (x_{32} \text{ } op_{32} \text{ } y_{32})$$

with $r_{64} = ((x_{32} + \epsilon_{x,prop}) \text{ } op_{64} \text{ } (y_{32} + \epsilon_{y,prop}))$, and $\epsilon_{x,prop}, \epsilon_{y,prop}$ being propagated rounding error for each operand.

B. Tracking Errors Per Code Line

To obtain the sequence of errors generated by one specific source line, FPChecker adds a second, more focused mechanism controlled by the environment variable `FPC_SAVE_LINE_ERRORS`. When this variable is set, the runtime allocates a small series manager indexed by source line and appends the observed error each time an instrumented operation mapped to that line executes. A resulting JSON file captures the temporal evolution of the selected line across loop iterations or repeated calls.

C. Gathering Statistics for Rounding Errors

Figure 2 illustrates our workflow for gathering statics about rounding error from FPChecker. Inputs to our workflow are the input interval, distribution function to sample values in the input interval and a kernel file, in this case dot product and long-summation. Outputs are computed p_k from collected local error using FPChecker, and s_k and d_k values, which are used to compute our modified bounds.

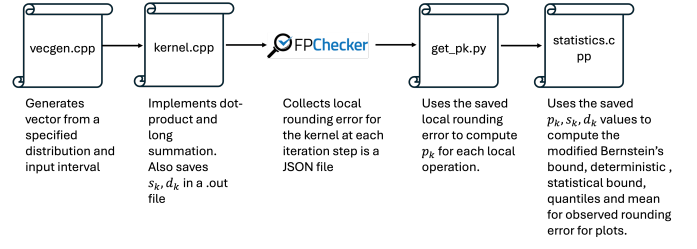


Fig. 2. Workflow for gathering key statistics from FPChecker

IV. ERROR MODEL

The standard δ error model for operation between two floating-point values a and b , $a \text{ } op_f \text{ } b = a \text{ } op_r \text{ } b(1 + \delta)$, where $op_r = \{+, -, *, /\}$ in exact real arithmetic, and op_f are floating-point counterparts of op_r . δ is the relative rounding error and its absolute value is bounded by the machine precision u , i.e. $|\delta| \leq u$. There are two widely used error models: (a) the deterministic model, where the error propagation assumes worst-case error bound, i.e., the rounding error $|a \text{ } op_f \text{ } b - a \text{ } op_r \text{ } b| \leq |a \text{ } op_r \text{ } b| u$; and (b) the statistical model [6], where $|a \text{ } op_f \text{ } b - a \text{ } op_r \text{ } b| \leq |a \text{ } op_r \text{ } b| r$ and r is drawn from some probability distribution with support interval $[-u, u]$. Commonly, r is drawn from a uniform distribution, i.e., $r \sim Unif(-u, u)$, and we will be using this distribution throughout this work. Both deterministic and standard statistical model fail to capture the structural floating-point properties or cases where floating-point operations are exact, as illustrated in Fig 1.

To remedy this issue, we propose a modified statistical error model, which models cases where floating-point operations are exact with some probability p_k . For instance, if we consider the floating-point addition with rounding error r , i.e., $a \oplus b = (a + b) + r$,

$$r \begin{cases} = 0 & \text{if } (a \oplus b) \text{ is exact} \\ \sim Unif(-u, u) & \text{if } (a \oplus b) \text{ is inexact} \end{cases}$$

We then use the mixture model [15] to model the above mentioned cases for rounding error as $r \sim p_k \delta_o + (1 - p_k) Unif(-u, u)$. Here, δ_o is a mass function with a huge spike when the rounding error is zero. Intuitively, the mixture error model states that for cases where the floating-point operation is exact, we sample r from the distribution δ_o with some probability p_k . For the cases where the floating-point operation is inexact, we sample r from a uniform distribution with support interval $[-u, u]$. Thus, p_k helps us capture the precise numerical behavior for a floating-point operation and inform the statistical model to obtain tighter error bounds. We compute p_k using the formula

$$p_k = \frac{\# \text{ samples with exact FP operation}}{\text{Total } \# \text{ samples}}$$

where $\# \text{ samples with exact FP operation}$ is obtained using the profiling information from FPChecker [13].

In this work, we will use the proposed mixture error model to obtain tighter error bounds for two numerical kernels—dot

product and long summation. These kernels are fundamental building blocks for much of linear algebra operations such as matrix-vector multiplication, axpy, etc.

A. Dot product

Dot product of two vectors $\vec{x}$ and $\vec{y}$ is defined by the inner product $\langle x, y \rangle$, which unrolls to the sum $S = \sum_k x_k y_k$. We can construct a chain of partial sums $s_1, s_2, \dots, s_N$, where the dot product S_N is the N^{th} partial sum s_N . Each partial sum s_k can be written as $s_k = s_{k-1} + x_k y_k$. Let us denote the floating-point k^{th} partial sum as $\hat{s}_k = fl(\hat{s}_{k-1} + \hat{d}_k)$, where $\hat{d}_k = fl(x_k y_k)$. Using relative error rounding error variables δ^m for multiplication operation and δ^a for addition operation, we can rewrite $\hat{d}_k = t_k(1 + \delta^m)$ and $\hat{s}_k = (\hat{s}_{k-1} + \hat{d}_k)(1 + \delta^a)$, where $t_k = x_k y_k$. Then,

$$\begin{aligned}\hat{s}_k &= (\hat{s}_{k-1} + t_k(1 + \delta_k^m))(1 + \delta_k^a) \\ &= \hat{s}_{k-1}(1 + \delta_k^a) + t_k(1 + \delta_k^m)(1 + \delta_k^a) \\ &= \hat{s}_{k-1} + (\hat{s}_{k-1} + t_k)\delta_k^a + t_k + t_k\delta_k^m + t_k\delta_k^a\delta_k^m.\end{aligned}$$

The local rounding error $\xi_k = \hat{s}_k - s_k$, will then be $\xi_k = t_k\delta_k^m + (\hat{s}_{k-1} + t_k)\delta_k^a + t_k\delta_k^a\delta_k^m$. Hence, the mixture of error model,

$$\begin{aligned}\xi_k &\sim p_k^m p_k^a \delta_o + (1 - p_k^m) p_k^a \text{Unif}(-\beta_k, \beta_k) \\ &\quad + p_k^m (1 - p_k^a) \text{Unif}(-\gamma_k, \gamma_k) \\ &\quad + (1 - p_k^m)(1 - p_k^a) (\text{Unif}(-\beta_k, \beta_k) \\ &\quad + \text{Unif}(-\gamma_k, \gamma_k) + u \text{Unif}(-\beta_k, \beta_k)).\end{aligned}$$

Here, $\beta_k \approx u|x_k y_k|$, $\gamma_k \approx u|\hat{s}_{k-1} + t_k|$ and $\delta^m = \delta^a = u$ since we are dealing with uniform precision in this work.

We can then compute the required parameters for the modified Bernstein bound in equation 4 as follows:

- Almost sure bound, $c_k := \max_{1 \leq k \leq n} r_k = \max_k (\beta_k + \gamma_k + u\beta_k)$.
- $\sum_{k=1}^N \text{Var}(\xi_k)$: For variance, define

$$X_k = t_k\delta_k^m; Y_k = (\hat{s}_{k-1} + t_k)\delta_k^a; Z_k = t_k\delta_k^m\delta_k^a.$$

Then, $\xi_k = X_k + Y_k + Z_k$. Variance of ξ_k ,

$$\begin{aligned}\text{Var}(\xi_k) &= \text{Var}(X_k) + \text{Var}(Y_k) + \text{Var}(Z_k) \\ &\quad + 2\text{Cov}(X_k, Y_k) + 2\text{Cov}(X_k, Z_k) \\ &\quad + 2\text{Cov}(Y_k, Z_k).\end{aligned}$$

$$\text{Var}(X_k) = (1 - p_k^m)\beta_k^2/3, \text{Var}(Y_k) = (1 - p_k^a)\gamma_k^2/3 \text{ and } \text{Var}(Z_k) = \frac{(1-p_k^a)}{3} \frac{(1-p_k^m)}{3} u^2 \beta_k^2.$$

Let,

$$\begin{aligned}V_{var} &= (1 - p_k^m)\beta_k^2/3 + (1 - p_k^a)\gamma_k^2/3 + \\ &\quad \frac{(1 - p_k^a)}{3} \frac{(1 - p_k^m)}{3} u^2 \beta_k^2.\end{aligned}$$

Hence,

$$\begin{aligned}V_{cov,k} &= V_{var} + 2\rho_k^{XY} \sqrt{\text{Var}(X_k)\text{Var}(Y_k)} \\ &\quad + 2\rho_k^{XZ} \sqrt{\text{Var}(X_k)\text{Var}(Z_k)} \\ &\quad + 2\rho_k^{YZ} \sqrt{\text{Var}(Y_k)\text{Var}(Z_k)}\end{aligned}$$

with $0 \leq \rho_k^{XY}, \rho_k^{XZ}, \rho_k^{YZ} \leq 1$. Covariance between multiplication and addition, i.e., XY term, only matter. This is because, $X_k \sim \mathcal{O}(u|t_k|)$ and $Y_k \sim \mathcal{O}(u|a_k|)$ are first order rounding-error terms, while $Z_k \sim \mathcal{O}(u^2|t_k|)$ is a second-order rounding-error term. Hence, $\text{Cov}(X_k, Y_k) \sim \mathcal{O}(u^2|t_k||a_k|)$, where as, $\text{Cov}(X_k, Z_k) \sim \mathcal{O}(u^3|t_k|^2)$ and $\text{Cov}(Y_k, Z_k) \sim \mathcal{O}(u^3|a_k||t_k|)$. Hence, XZ and YZ terms are smaller than XY term by roughly a factor of u . Hence,

$$V_{cov,k} = V_{var} + 2\frac{\rho_k}{3} \sqrt{(1 - p_k^m)\beta_k^2(1 - p_k^a)\gamma_k^2}.$$

Total variance, $V(E_N) = \sum_{k=1}^N \text{Var}(\xi_k) + 2\sum_{i < j} \text{Cov}(\xi_i, \xi_j)$. But Bernstein's inequality assumes independence between summands. Hence

$$\begin{aligned}\text{Var}(E_N) &= \sum_{k=1}^N \text{Var}(\xi_k) \\ &= \sum_{k=1}^N (1 - p_k^m)\beta_k^2/3 + (1 - p_k^a)\gamma_k^2/3 + \\ &\quad \frac{(1 - p_k^a)}{3} \frac{(1 - p_k^m)}{3} u^2 \beta_k^2 + \\ &\quad \frac{2}{3} \rho_k \sqrt{(1 - p_k^m)(1 - p_k^a)\beta_k\gamma_k}\end{aligned}\quad (5)$$

B. Long summation

Given a vector $\vec{x}$, the long summation $S = \sum_k x_k$. Similar to the dot product case, we can construct a partial sum, $\hat{s}_k = fl(\hat{s}_{k-1} + x_k)$. Using the relative error rounding variable δ^a , we can write $\hat{s}_k$ as $\hat{s}_k = (\hat{s}_{k-1} + x_k)(1 + \delta_k^a)$. The local rounding error, $\xi_k = \hat{s}_k - s_k$ will then be $\xi_k = (\hat{s}_{k-1} + x_k)\delta_k^a$. Hence, the mixture of error model,

$$\xi_k \sim p_k^a \delta_o + (1 - p_k^a) \text{Unif}(-\gamma_k, \gamma_k).$$

Here, $\gamma_k \approx u|\hat{s}_{k-1} + x_k|$. We can then compute the required parameters for the modified Bernstein bound in equation 4 as follows:

- Almost sure bound, $c_k := \max_{1 \leq k \leq n} r_k = \max_k (\gamma_k)$.
- Variance, $\sum_{k=1}^N \text{Var}(\xi_k)$:

$$\text{Var}(\xi_k) = (1 - p_k^a) \text{Var}(\text{Unif}(-\gamma_k, \gamma_k)).$$

For a uniform distribution over an interval $[a, b]$, variance of the distribution is given by $(b - a)^2/12$. Hence,

$$\text{Var}(\xi_k) = (1 - p_k^a) \text{Var}(\text{Unif}(-\gamma_k, \gamma_k)) = (1 - p_k^a) \gamma_k^2/3.$$

Since Bernstein's inequality assumes independence of summands, $\sum_{k=1}^N \text{Var}(\xi_k) = \sum_{k=1}^N (1 - p_k^a) \gamma_k^2/3$.

C. Computing d in modified Bernstein's inequality 4.

In our case of sequential inner product, errors propagate down the chain: a rounding error at step j permanently changes the value of the accumulator, which directly alters the grid spacing and thus the error distribution for all future steps $i > j$. Let each rounding error ξ_k be a vertex in a graph G . An edge exists between ξ_i and ξ_j if they are statistically dependent $\text{Cov}(\xi_j, \xi_i) \neq 0$. Because ξ_j depends on the accumulator $\hat{s}_{j-1}$

which contains the history of all past errors - ξ_j is dependent on $\xi_1, \xi_2, \dots, \xi_{j-1}$. Hence, the dependency graph is a complete directed graph (clique). Normally, dependency graph inequalities require a small maximum degree d . Because our graph is a clique, the maximum degree is $d = N - 1$. To make the dependency approach useful for average-case certificates, we look at the strength of the dependencies. Looking at our error model $\xi_i \approx u(\hat{s}_{i-1} + a_i b_i)$ and $\xi_i \approx u(u(\hat{s}_{i-2} + a_{i-1} b_{i-1}) + a_i b_i)$. Hence $Cov(\xi_j, \xi_i) \approx \mathcal{O}(u^{i-j})$. Because the u is incredibly small, the dependency decays exponentially fast relative to the step distance $i - j$. We can thus construct a u -dependency graph by drawing an edge only if $|Cov(\xi_j, \xi_i)| > u^2$. Since $Cov(\xi_i, \xi_{i-1}) \propto u$ and $Cov(\xi_i, \xi_{i-2}) \propto u^2$, any step distance greater than 1 can be treated as independent for the first-order bound. This simplifies our dense clique into a line graph where each node ξ_k only connects to its immediate neighbors ξ_{k-1} and ξ_{k+1} . Thus, the maximum degree of dependency that we use in our bounds, $d = 2$.

V. EXPERIMENTAL EVALUATION

In this section, we compare the error bounds for dot product and long summation derived in Section IV against the deterministic error bounds and standard statistical bounds derived by Higham and Mary. For long summation, deterministic error bound is given by $((1 + u)^n - 1) \sum_k |x_k|$, which is $\mathcal{O}(nu)$. Higham and Mary's statistical bound [6] is given by $\gamma \sqrt{nu} \sum_k |x_k|$, and is of order, $\mathcal{O}(\sqrt{nu})$. Bhola and Duraisamy [3] derive the value of γ as $\frac{1}{1-u} \sqrt{2 \log(\frac{2}{1-\zeta})}$, where $\zeta \in [0, 1)$ is the confidence interval value. For the dot product, deterministic error bound is $\approx ((1 + u)^n - 1) \sum_k |x_k y_k|$ and standard statistical bound is $\approx \gamma \sqrt{nu} \sum_k |x_k y_k|$. We plot the error bounds for two intervals $[1, 2]$ and $[1, 5]$. For each interval, we perform two experiments - in the first experiment, each element of the second vector has same sign and in the second experiment, elements of the second vector have alternating signs, i.e.,

```
for (int i = 0; i < N; i++) {
    float b = dist(gen);
    // float b = 1.0f;
    // For long summation,
    the second vector just has +1 and -1.
    if (i%2 == 0)
        csvfile << b << ",_";
    else
        csvfile << -b << ",_"; }
```

The alternating signs capture the case where the probability of cancellations or exactness of the subtraction operation (e.g., due to Sterbenz lemma [2]) is high. This can be visualized using the p_k plots for both the cases.

Figure 3 plots p_k values for dot product kernel at the 100th partial sum for alternating sign case, and Figure 4 plots p_k values for dot product kernel at the 100th partial sum for the same sign case, i.e., all elements of the second vector have same sign. The oscillatory nature of Figure 3 demonstrates the effect of cancellation of errors, i.e., for odd k values,

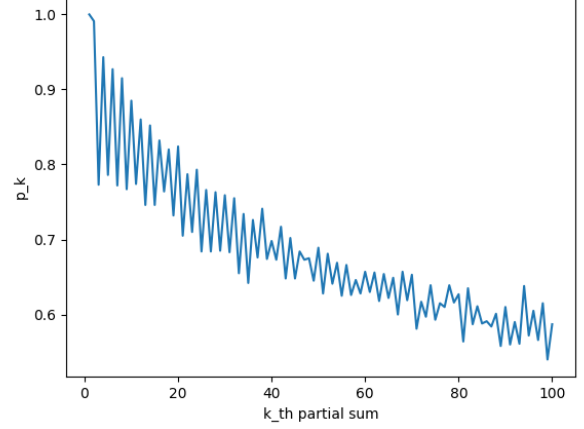


Fig. 3. p_k plot for dot product computation for the alternating sign case for interval $[1, 2]$. p_k is the probability with which an FP operation is exact.

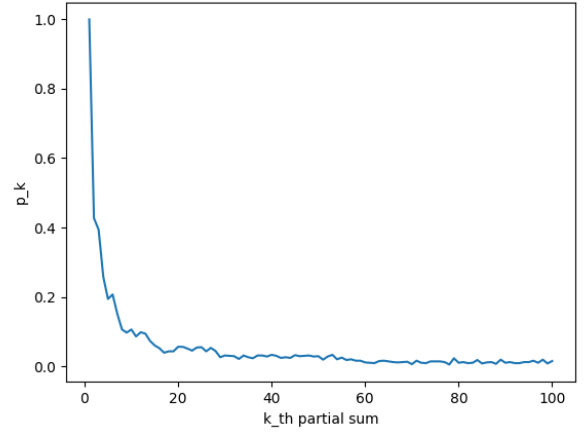


Fig. 4. p_k plot for dot product computation for the same sign case for interval $[1, 2]$.

where the addition is between numbers of opposite signs, the probability of exactness of $\oplus$ is higher than for even k values. Furthermore, the oscillatory nature is more prominent for early k values because the accumulator $\hat{s}_{k-1} \approx x_k$. However, for later k values, as the accumulator $\hat{s}_{k-1}$ becomes larger than x_k , probability of such cancellations reduces and the cancellations become less periodic. In figure 4, p_k values decrease exponentially to 0 after only a few k values, since the cancellations of errors or exactness of $\oplus$ operation between two values of similar magnitude does not take place more often due to the values being of same sign. Hence, Sterbenz lemma, which was a major source of exactness in the first case, does not apply for this case.

Furthermore, the choice of intervals $[1, 2]$ and $[1, 5]$ is motivated by the fact that the effect of such cancellation of errors or exactness of operation is expected to be more prominent for narrower interval, where the sampled vector values are close to each other and more likely to be within the Sterbenz range. This is validated by Figures 5 and 6. Figure 5 is less

oscillatory than Figure 3. Furthermore, p_k values for interval $[1, 5]$ is lower than the p_k value for interval $[1, 2]$.

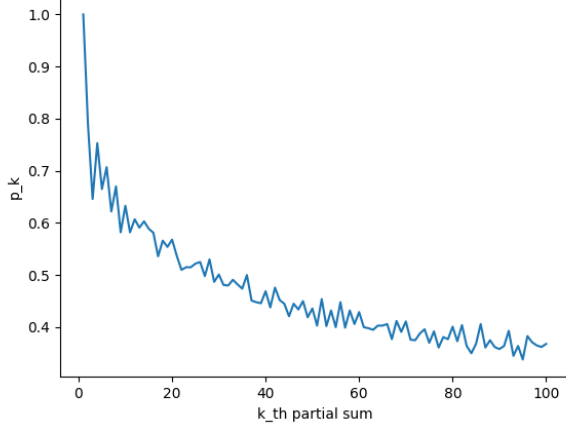


Fig. 5. p_k plot for dot product computation for the alternating sign case for interval $[1, 5]$.

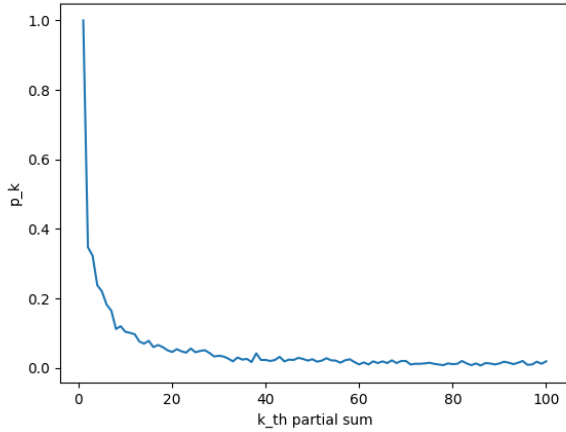


Fig. 6. p_k plot for dot product computation for the same sign case for interval $[1, 5]$.

We next plot the error bounds for long summation and dot product numerical kernels.

A. Long summation

Figure 7 and 8 plot error bounds for alternating sign and same sign case for interval $[1, 2]$ respectively. The green plot illustrates the growth of deterministic error bound with vector size. The red plot illustrates the growth of Higham and Mary statistical bound with vector size. The purple plot illustrates the growth of our modified/structural statistical bound with vector size. The blue plot is the mean of the actual error distribution for different vector sizes and the orange plot is the 95th quantile of the actual error distribution. *Since Bernstein's bound is a tail bound which bounds the tail behavior of the actual error distribution, we expect that a sound 95%*

high probability/Bernstein's bound would be greater than the 95th quantile of the actual error distribution. All our plots demonstrate that our derived bound is sound.

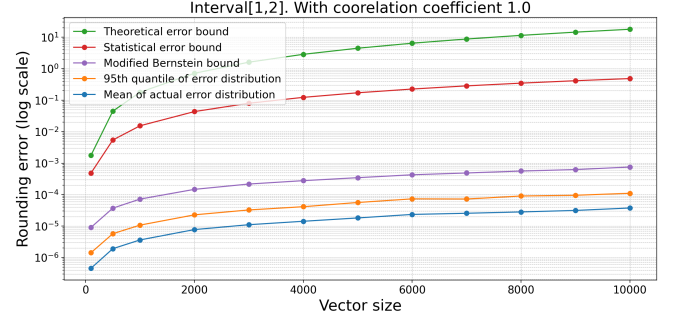


Fig. 7. Error bounds for long summation, $\sum_k x_k$ for alternating sign case for interval $[1, 2]$

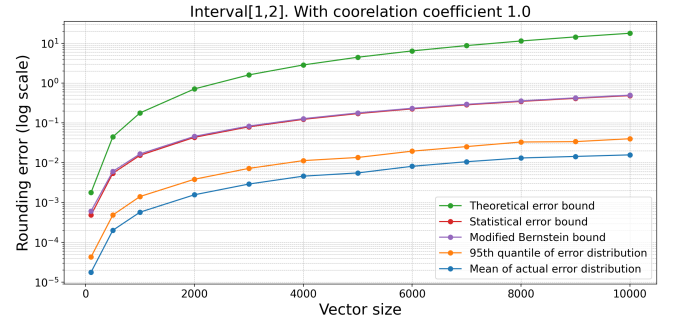


Fig. 8. Error bounds for long summation, $\sum_k x_k$ for same sign case for interval $[1, 2]$

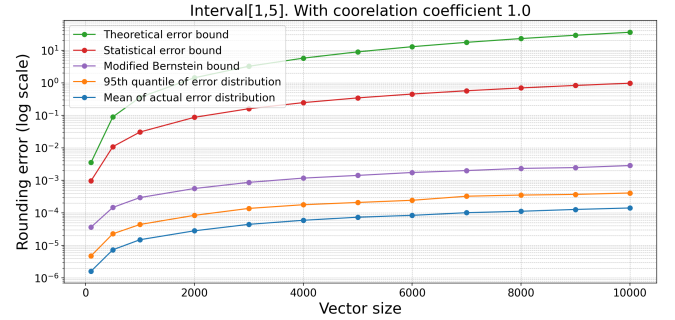


Fig. 9. Error bounds for long summation, $\sum_k x_k$ for alternating sign case for interval $[1, 5]$

Futhermore, using an order of magnitude comparison for the bounds, we can see that the deterministic bound ($\mathcal{O}(10^1)$ for $n = 10,000$) for alternating sign case is roughly six orders of magnitude greater than the actual rounding error ($\mathcal{O}(10^{-5})$ for $n = 10,000$). The standard statistical error bound ($\mathcal{O}(10^{-1})$ for $n = 10,000$) is roughly four orders of magnitude greater than the actual rounding error ($\mathcal{O}(10^{-5})$ for $n = 10,000$). *Our structural statistical bound ($\mathcal{O}(10^{-4})$ for $n = 10,000$) on the other hand is roughly one order of magnitude greater than the actual rounding rounding error ($\mathcal{O}(10^{-5})$ for $n = 10,000$).*

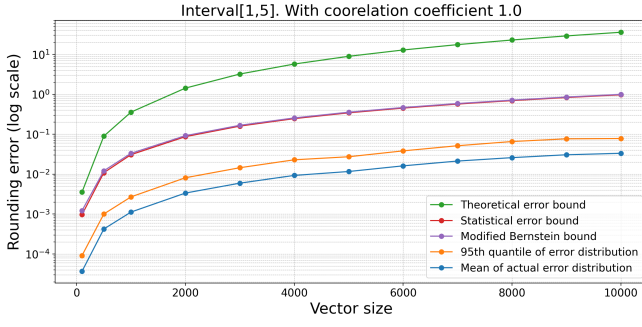


Fig. 10. Error bounds for long summation, $\sum_k x_k$ for same sign case for interval $[1, 5]$

Thus, our structural bound captures the actual numerical behavior by virtue of p_k values, and provides tighter error bounds.

For the same sign case (Figure 8 and 10), where we do not anticipate much cancellation of errors or exactness of operations, our structural bound is closer to the standard statistical bound. This shows that our error model faithfully captures the non-linearity and local error correlations, and is not an oversimplification.

For interval $[1, 5]$, our structural bound ($\mathcal{O}(10^{-3})$ for $n = 10,000$) is roughly two orders of magnitude greater than the actual rounding error ($\mathcal{O}(10^{-5})$ for $n = 10,000$). This is expected since p_k values for $[1, 5]$ is lower than $[1, 2]$, pushing the effective n_{eff} to a higher value. Here, n_{eff} is the effective number of operations contributing to the rounding error accumulation. For standard statistical bounds, $n_{\text{eff}} = n$, since this bound assumes that all operations contribute to the rounding error. However, in our structural statistical bound $n_{\text{eff}} \approx \sum_k (1 - p_k^a) < n$.

B. Dot product

Figure 11, 12, 13 and 14, show the error scaling plots for alternating and same sign case for intervals $[1, 2]$ and $[1, 5]$, respectively.

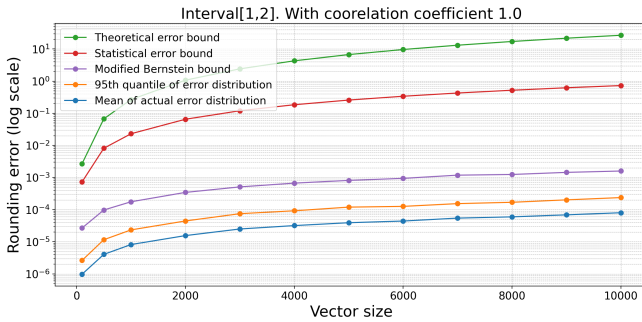


Fig. 11. Error bounds for dot product, $\sum_k x_k y_k$ for alternating sign case for interval $[1, 2]$

Using an order of magnitude comparison for bounds in interval $[1, 2]$, we can see that the deterministic bound ($\mathcal{O}(10^1)$ for $n = 10,000$) for alternating sign case is roughly six orders of magnitude greater than the actual rounding error

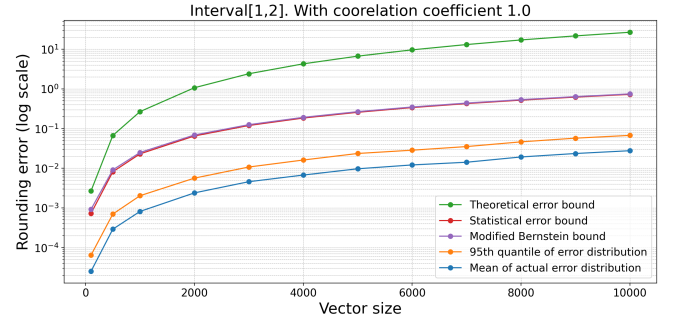


Fig. 12. Error bounds for dot product, $\sum_k x_k y_k$ for same sign case for interval $[1, 2]$

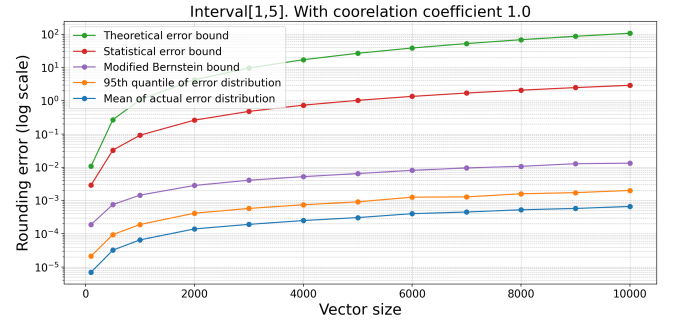


Fig. 13. Error bounds for dot product, $\sum_k x_k y_k$ for alternating sign case for interval $[1, 5]$

($\mathcal{O}(10^{-5})$ for $n = 10,000$). The standard statistical error bound ($\mathcal{O}(10^{-1})$ for $n = 10,000$) is roughly four orders of magnitude greater than the actual rounding error ($\mathcal{O}(10^{-5})$ for $n = 10,000$). Our structural statistical bound ($\mathcal{O}(10^{-3})$ for $n = 10,000$) on the other hand is roughly two order of magnitude greater than the actual rounding rounding error ($\mathcal{O}(10^{-5})$ for $n = 10,000$). Thus, our structural bound captures the actual numerical behavior by virtue of p_k values, and provides tighter error bounds. We also see a similar order of magnitude comparison for the interval $[1, 5]$. For the same sign case, our structural bound is closer to the standard statistical bound, as was the case for long summation. Furthermore, our error bound (purple plot) is greater than the 95th quantile (orange plot) of the actual error distribution, which shows that our tight error bound for dot product is sound.

VI. TESTING OUR MODIFIED BERNSTEIN-BOUND ON A PHYSICAL INTERVAL

Consider the numerical detection of a weak density mode in a periodic compressible flow, $\rho(x) = \rho_o(1 + \epsilon \cos(kx))$, $|\epsilon| \leq \epsilon_{\text{max}}$. Compressible-flow simulations routinely track small density Fourier modes because they represent acoustic waves, pressure-density perturbations, and compressibility effects, which are important in aeroacoustics, compressible turbulence, combustion/reactive flows, etc. Accurately resolving these density-wave amplitudes is therefore essential for distinguish-

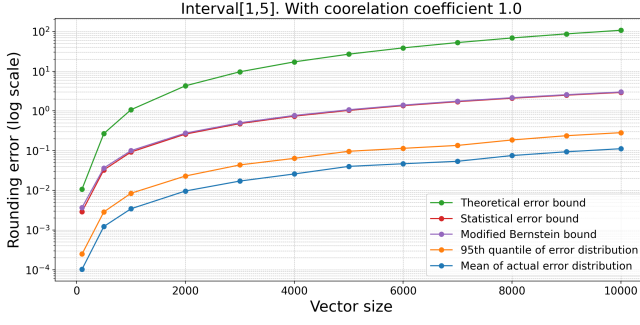


Fig. 14. Error bounds for dot product, $\sum_k x_k y_k$ for same sign case for interval $[1, 5]$

ing physically meaningful acoustic/compressible modes from numerical noise.

To detect the density wave, the Fourier coefficient, $A_k = \frac{2}{L} \int_0^L \rho(x) \cos(kx) dx$, is computed. This analytical integration is computed numerically using the quadrature sum, $A_{k,N} = \frac{2}{L} \sum_{j=0}^{N-1} w_j \rho(x_j) \cos(kx_j)$. The numerical kernel for this quadrature sum is the long-summation, and the floating-point integration error is given by the rounding error for this long-summation kernel. This is cancellation-heavy because the term $w_j \rho(x_j) \cos(kx_j)$ is oscillatory, i.e., positive and negative, making it an ideal case study for demonstrating effectiveness of our derived error bound. Substituting the value of $\rho(x)$ in $w_j \rho(x_j) \cos(kx_j)$, we get the summand $t = w_j \rho_o (1 + \epsilon \cos(kx_j)) \cos(kx_j)$. It can be easily observed

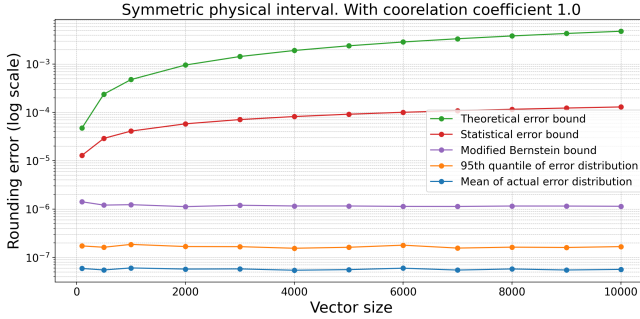


Fig. 15. Error bounds for long-summation for the physical interval.

that $t_{min} = -w_j \rho_o (1 + \epsilon_{max})$ and $t_{max} = w_j \rho_o (1 + \epsilon_{max})$. Thus, the interval is symmetric, as is the case for many physical problems. Since, t is sinusoidal and the points are clustered around the end-points of the interval, we do not sample points using a uniform distribution directly on the interval $[t_{min}, t_{max}]$. Rather, we sample $\theta = kx_j \sim \text{Unif}[0, 2\pi]$, and then substitute the sampled θ in t to get the summands for the long-summation. We assign equal quadrature weights to all points $w_j = L/N$, and we vary N as illustrated in Fig 15. We fix $\rho_o = 1$ and $\epsilon_{max} = 10^{-3}$. We then use FPChecker to collect p_k values for the long summation kernel on this interval, shown in Fig 16, and compute our modified Bernstein's bound by varying N , as before. As before, our bound is within two-orders of magnitude of the mean of the

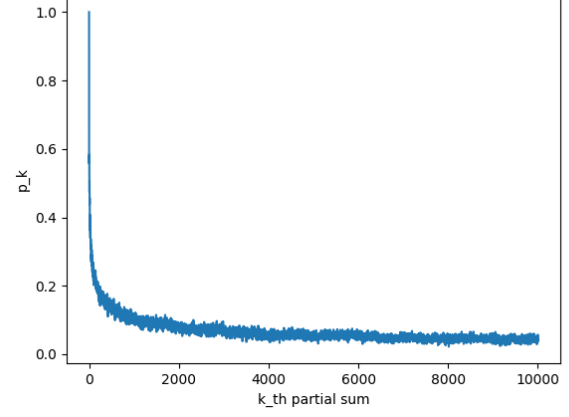


Fig. 16. p_k values for long summation computed on the physical interval.

actual rounding error, and follows the behavior of the actual rounding-error closely. For this particular example, the analytical integration $A_k = \rho_o \epsilon = 10^{-3}$. From figure 15 we observe that the deterministic error bound is greater than the true solution for larger N , and even the standard statistical bound is comparable to the true solution for larger N . However, our modified Bernstein-bound which captures the cancellation behavior using p_k values, is near constant even for larger N , and is sound physically.

VII. CONCLUSION AND FUTURE WORK

In this work, we derive tighter error bounds for long summation and dot product kernels using probabilistic error analysis. We introduce the p_k parameter as a way of capturing the exactness of floating-point operation. We show that for cancellation heavy case (alternating signs), the p_k parameter effectively captures the numerical behavior and its incorporation into Bernstein's tail bound provide tighter error estimation. Our bound is roughly within two orders of magnitude of the actual rounding error for both long summation and dot product kernels. We further provide a derivation of modified Bernstein's bound using Janson's dependency graph approach to capture weak correlations between local error to prevent over simplification of our error model. Finally, we show the applicability of the derived bounds on a physical example of computing the magnitude of Fourier modes for density wave function used in compressible-flow simulations. In the immediate future, we will extend this approach to larger numerical kernels used in linear solvers from BLAS or PolyBench packages, and provide an error growth estimate for these linear solvers. Our eventual goal is to provide tighter error bounds and accurate characterization of numerical behavior in solver codes such as the conjugate gradient method.

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